

LAB EXPERIMENT – PART 2

Performance Analysis Using Type-2 Hypervisor

VMware Workstation

CPU Performance Analysis of an Ubuntu Virtual Machine using Sysbench

Student Information

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1. Aim

To create and verify a virtual machine on the VMware Workstation Type-2 hypervisor, inspect its CPU, memory and disk configuration, monitor resource utilization, and measure CPU performance using Sysbench.

2. Experimental Configuration

Configured / Observed Value	Parameter
VMware Workstation (Type-2)	Hypervisor
Ubuntu 22.04.5 LTS	Guest OS
1 processor, 2 cores = 2 vCPU	CPU allocation
13th Gen Intel Core i5-13450HX presented to guest	CPU model
2048 MB (approximately 2 GB)	Memory allocation
20 GB virtual disk	Virtual disk
NAT	Network configuration
VMware / full virtualization	Virtualization information

3. Procedure

- Launch VMware Workstation and select Create a New Virtual Machine.
- Choose the Typical configuration and select the Ubuntu ISO image.
- Create the VM and configure a 20 GB virtual disk.
- Customize hardware to allocate 2 GB RAM and 2 vCPU; configure the network adapter using NAT.
- Power on the VM, install Ubuntu, restart, and log in.
- Verify the VM using `hostnamectl`, `lscpu`, `free -h` and `df -h`.
- Run `top` to observe CPU utilization, memory utilization, processes and load average.
- Install Sysbench and execute `sysbench cpu --cpu-max-prime=20000 run`.
- Record execution time, total events, events per second and latency statistics.
- Shut down the Ubuntu VM using `sudo poweroff` after completing the measurements.

4. Commands Used

```
hostnamectl
lscpu
free -h
df -h
top
sudo apt update
sudo apt install sysbench -y
sysbench --version
sysbench cpu --cpu-max-prime=20000 run
sudo poweroff
```

5. Observed Results

Observed Result	Metric
Ubuntu 22.04.5 LTS	Operating system
x86_64	Architecture
2 CPUs / 2 cores	Virtual CPUs
13th Gen Intel Core i5-13450HX	CPU model reported by guest
VMware; full virtualization	Virtualization
Approximately 1.9 GiB total	Memory reported by free -h
20 GB virtual disk	Disk
10.0004 s	Sysbench execution time
14,410	Sysbench total events
1,440.80	Sysbench events per second
0.65 ms	Minimum latency
0.69 ms	Average latency
1.89 ms	Maximum latency

6. Resource Utilization Observations

- The guest reports 2 online CPUs and approximately 2 GB of allocated memory.
- The free -h output shows approximately 1.9 GiB total memory, with available memory around 865 MiB at the captured instant.
- The df -h output shows a 20 GB virtual root filesystem, with approximately 12 GB used and 6.5 GB available.
- The top display provides live CPU, memory, process and load information inside the Ubuntu guest.
- Because VMware Workstation is a Type-2 hypervisor, the guest's virtual resources are mediated through the host operating system.

7. Result and Conclusion

Part 2 was completed using VMware Workstation as a Type-2 hypervisor. The Ubuntu virtual machine was configured with 2 vCPUs, 2 GB RAM and a 20 GB virtual disk. The VM configuration was verified using standard Linux system commands, and guest resource utilization was observed using top.

For the supplied Sysbench CPU run, the VM completed the benchmark in 10.0004 seconds and processed 14,410 events at 1,440.80 events per second, with an average latency of 0.69 ms. These values form the Type-2 result set for comparison with the Type-1 Proxmox VE experiment.

8. Type-2 Hypervisor Interpretation

- Architecture: VMware Workstation operates above the host operating system, while Ubuntu runs as the guest OS.
- Host–guest interaction: CPU, memory, storage and networking are presented to Ubuntu as virtual resources and are mediated by the virtualization layer and host.
- Isolation: The guest operates on a virtual disk and virtual CPU resources rather than directly controlling the host's physical devices.
- Typical use cases: Type-2 virtualization is suitable for desktop virtualization, development/testing, laboratory work and running another operating system on a workstation.
- Performance considerations: Benchmark throughput and latency can be affected by virtualization overhead, host workload, resource contention and the configuration of the virtual machine.

9. Final Observation Record

Part 2 Result	Parameter
VMware Workstation	Hypervisor
Type-2	Hypervisor type
Ubuntu 22.04.5 LTS	Guest OS
2 vCPU	CPU
2 GB	Memory
20 GB	Disk
10.0004 s	Total execution time
14,410	Total events
1,440.80	Events per second
0.69 ms	Average latency

10. Screenshots / Experimental Evidence

The supplied screenshots are included below in the same evidence-oriented format as the Type-1 reference. Each figure is placed before its caption, with the caption immediately beneath the corresponding image.

```
nupur@nupur-virtual-machine:~$ hostnamectl
Static hostname: nupur-virtual-machine
Icon name: computer-vm
Chassis: vm
Machine ID: 80a2e65ecdabc4d41a7ed2270adacbab7
Boot ID: 1ac7a15768444772a855f3459d94be09
Virtualization: vmware
Operating System: Ubuntu 22.04.5 LTS
Kernel: Linux 6.8.0-138-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
```

Figure 1. Ubuntu VM: hostnamectl output showing the hostname, Ubuntu operating system, kernel, architecture and VMware virtual hardware information.

```
nupur@nupur-virtual-machine:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 45 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13450HX
CPU family: 6
Model: 183
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Stepping: 1
BogoMIPS: 5222.39
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpeibg rdtscp lm constant_tsc arch_perfmon rep_good nopl xt
opology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dno
wprefetch pti ssbd ibrs lbrs l1dbs stlbp fsgsbase tsc_adjust bti1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni arat uml
p gfnl vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_l1d arch_capabilities

Virtualization features:
Hypervisor vendors: VMware
Virtualization type: full
Caches (sum of all):
L1d: 96 KIB (2 instances)
L1i: 64 KIB (2 instances)
L2: 2.5 MiB (2 instances)
L3: 20 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1
Vulnerabilities:
Gather data sampling: Not affected
Indirect target selection: Mitigation: Aligned branch/return thunks
Itlb multihit: Not affected
L1tf: Mitigation; PTE Inversion
Mds: Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown: Mitigation; PTI
Mmio stale data: Unknown: No mitigations
Reg file data sampling: Vulnerable: No microcode
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; IBRS; IBPB conditional; STIBP disabled; RSB filling; PBRSSB-eIBRS Not affected; BHI SW loop, KVM SW loop
Srbds: Not affected
Taa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
```

Figure 2. Ubuntu VM: lscpu output showing x86_64 architecture, 2 CPUs/cores, the presented Intel CPU model and VMware full virtualization.

```
nupur@nupur-virtual-machine:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	869Mi	256Mi	23Mi	794Mi	865Mi
Swap:	2.1Gi	78Mi	2.0Gi			

Figure 3. Ubuntu VM: free -h output showing memory and swap allocation/usage.

```
nupur@nupur-virtual-machine:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	193M	4.2M	188M	3%	/run
/dev/sda3	20G	12G	6.5G	64%	/
tmpfs	961M	0	961M	0%	/dev/shm
tmpfs	5.0M	4.0K	5.0M	1%	/run/lock
/dev/sda2	512M	6.1M	506M	2%	/boot/efi
tmpfs	193M	164K	192M	1%	/run/user/1000
/dev/sr0	157M	157M	0	100%	/media/nupur/CDROM

Figure 4. Ubuntu VM: df -h output showing the 20 GB virtual filesystem, used space and available space.

```
nupur@nupur-virtual-machine:~$ top
```

top - 10:22:18 up 15 min, 1 user, load average: 0.00, 0.03, 0.05
Tasks: 292 total, 2 running, 290 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.2 us, 0.8 sy, 0.0 ni, 97.7 id, 0.0 wa, 0.0 hi, 0.3 st, 0.0 st
MiB Mem : 1920.2 total, 258.1 free, 867.1 used, 795.1 buff/cache
MiB Swap: 2140.0 total, 2061.3 free, 78.7 used, 867.8 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1767	nupur	20	0	4436416	192400	76736	S	2.0	9.8	0:14.41	gnome-shell
2722	nupur	20	0	560236	58148	45092	S	1.0	3.0	0:02.31	gnome-terminal-
3829	nupur	20	0	13336	4336	3368	R	0.7	0.2	0:00.03	top
570	systemd+	20	0	14836	6140	6012	S	0.3	0.3	0:01.60	systemd-oomd
744	root	20	0	169920	8796	7356	S	0.3	0.4	0:02.56	vmtoolsd
1	root	20	0	166684	11860	8396	S	0.0	0.6	0:01.24	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.17	kworker/0:0H-kblockd
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0.0	0.0	0:00.04	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:00.24	rcu_preempt
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
23	root	rt	0	0	0	0	S	0.0	0.0	0:00.21	migration/1
24	root	20	0	0	0	0	S	0.0	0.0	0:00.05	ksoftirqd/1
25	root	20	0	0	0	0	I	0.0	0.0	0:00.37	kworker/1:0-rcu_par_gp
26	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
27	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
28	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_
29	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
30	root	20	0	0	0	0	I	0.0	0.0	0:00.10	kworker/0:2-inet_frag_wq
31	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
35	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-write
36	root	20	0	0	0	0	S	0.0	0.0	0:00.12	kcompactd0
37	root	25	5	0	0	0	S	0.0	0.0	0:00.00	ksmd
39	root	20	0	0	0	0	I	0.0	0.0	0:00.50	kworker/1:2-rcu_par_gp
40	root	39	19	0	0	0	S	0.0	0.0	0:00.00	khugepaged
41	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kinte
42	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kbloc
43	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-blkcg
44	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	irq/9-acpi
45	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-top_d

Figure 5. Ubuntu VM: top output showing CPU utilization, memory utilization, load average and active processes.

```
nupur@nupur-virtual-machine:~$ sudo apt install sysbench -y
```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
libb101 liblua5.1-2 liblua5.1-common libmysqlclient21 libpq5
mysql-common
The following NEW packages will be installed:
libb101 liblua5.1-2 liblua5.1-common libmysqlclient21 libpq5
mysql-common sysbench
8 upgraded, 7 newly installed, 0 to remove and 460 not upgraded.
Need to get 1,930 kB of archives.
After this operation, 8,548 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 libb101 amd64 0.3.112-1build1 [7,176 B]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 liblua5.1-2 amd64 2.1.0-beta3+dfsg-6ubuntu0.1 [46.6 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 liblua5.1-2 amd64 2.1.0-beta3+dfsg-6ubuntu0.1 [256 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7,112 B]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libmysqlclient21 amd64 8.0.46-0ubuntu0.22.04.4 [1,336 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.24-0ubuntu0.22.04.1 [157 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 sysbench amd64 1.0.20+ds-2 [120 kB]
Fetched 1,930 kB in 3s (488 kB/s)
Selecting previously unselected package libb101:amd64.
(Reading database ... 201775 files and directories currently installed.)
Preparing to unpack .../0-libb101_0.3.112-1build1_amd64.deb ...
Unpacking libb101:amd64 (0.3.112-1build1) ...
Selecting previously unselected package liblua5.1-common.
Preparing to unpack .../1-liblua5.1-common_2.1.0-beta3+dfsg-6ubuntu0.1_all.d
eb ...
Unpacking liblua5.1-common (2.1.0-beta3+dfsg-6ubuntu0.1) ...
Selecting previously unselected package liblua5.1-2:amd64.
Preparing to unpack .../2-liblua5.1-2_2.1.0-beta3+dfsg-6ubuntu0.1_amd64.deb
...
Unpacking liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6ubuntu0.1) ...
Selecting previously unselected package mysql-common.
Preparing to unpack .../3-mysql-common_5.8+1.0.8_all.deb ...
Unpacking mysql-common (5.8+1.0.8) ...
Selecting previously unselected package libmysqlclient21:amd64.
Preparing to unpack .../4-libmysqlclient21_8.0.46-0ubuntu0.22.04.4_amd64.deb ...
Unpacking libmysqlclient21:amd64 (8.0.46-0ubuntu0.22.04.4) ...
Selecting previously unselected package libpq5:amd64.
Preparing to unpack .../5-libpq5_14.24-0ubuntu0.22.04.1_amd64.deb ...
Unpacking libpq5:amd64 (14.24-0ubuntu0.22.04.1) ...
Selecting previously unselected package sysbench.
Preparing to unpack .../6-sysbench_1.0.20+ds-2_amd64.deb ...
Unpacking sysbench (1.0.20+ds-2) ...
Setting up mysql-common (5.8+1.0.8) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.c
nf (my.cnf) in auto mode
Setting up libmysqlclient21:amd64 (8.0.46-0ubuntu0.22.04.4) ...
Setting up libpq5:amd64 (14.24-0ubuntu0.22.04.1) ...
Setting up liblua5.1-common (2.1.0-beta3+dfsg-6ubuntu0.1) ...
Setting up libb101:amd64 (0.3.112-1build1) ...
Setting up liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6ubuntu0.1) ...
Setting up sysbench (1.0.20+ds-2) ...
Processing triggers for man-db (2.10.2-1) ...

Figure 6. Ubuntu VM: Sysbench installation output showing package installation and configuration.

```

nupur@nupur-virtual-machine:~$ sysbench --version
sysbench 1.0.20
nupur@nupur-virtual-machine:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second: 1440.80

General statistics:
  total time:          10.0004s
  total number of events: 14410

Latency (ms):
  min:                 0.65
  avg:                 0.69
  max:                 1.89
  95th percentile:    0.90
  sum:                 9994.20

Threads fairness:
  events (avg/stddev): 14410.0000/0.00
  execution time (avg/stddev): 9.9942/0.00

```

Figure 7. Ubuntu VM: Sysbench CPU benchmark output for --cpu-max-prime=20000, showing execution time, events, throughput and latency.

11. Performance Analysis Summary

The final benchmark screenshot records 1,440.80 events per second, 14,410 total events and 10.0004 seconds of execution time. Minimum, average and maximum latency are 0.65 ms, 0.69 ms and 1.89 ms respectively.

12. Key Observations

- The guest successfully recognizes VMware as its virtualization environment.
- The VM exposes 2 virtual CPUs and approximately 2 GB of memory to Ubuntu.
- The guest sees a 20 GB virtual disk through the virtual storage layer.
- The Type-2 arrangement introduces an additional software layer between the guest and physical hardware, which can influence performance.
- The Sysbench result provides a reproducible CPU-performance measurement for comparison with the Type-1 result.